AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I217*

Team Name: *Brute Force*

Domain: *7. Digital Technologies, AI and Cybersecurity*

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Presentation Outline

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Problem Statement

Due to inefficient navigation systems inside the campus, freshers, parents, and visitors find difficulty locating places within complex grounds and huge buildings like AB1. Furthermore, students struggle to track real-time shuttle buses, leading to chronic campus disorientation and missed academic lectures.

Problem Understanding

- Target Users: Freshers, visiting parents, guests, and regular students, cab drivers, delivery persons, etc...
- Root Causes: Complete GPS signal attenuation inside concrete blocks, lack of centralized real-time transit telemetry and lesser navigation for multi storey buildings.
- Consequences: Severe navigational friction, delayed arrivals, visitor frustration, and missed lectures due to unfamiliar venue changes.

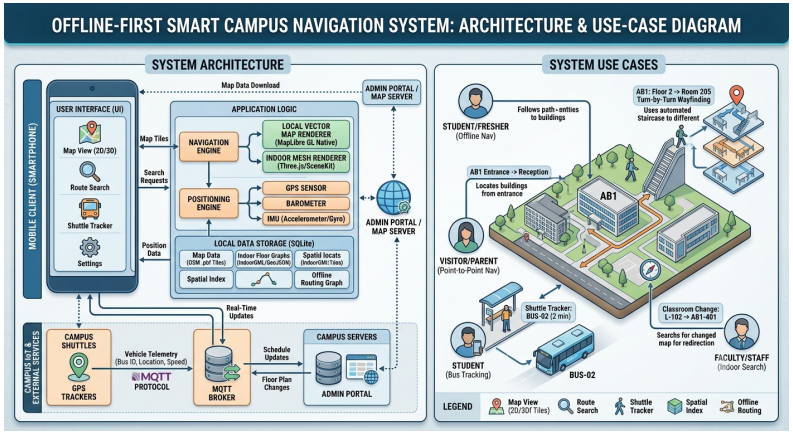
Existing Gaps & Challenges

- Standard commercial mapping tools fail entirely indoors and inside dense concrete structures due to satellite blockage.
- Existing transit solutions lack lightweight real-time telemetry, causing high battery drain and failure in intermittent connectivity zones.
- Unavailability of offline spatial packages forces complete dependence on unstable mobile data networks across campus dead zones.

Our Proposed Solution

- Offline-First Smart Campus Navigation System combining high-performance 2D vector maps and 3D indoor multi-floor models.
- Integrates lightweight MQTT telemetry for real-time shuttle tracking and smartphone sensor fusion for seamless indoor-outdoor positioning.
- Delivers instant on-device pathfinding and zero-internet reliability, eliminating navigation friction for all campus stakeholders.

Technical Approach



System parses offline OSM .pbf files via MapLibre, fuses GPS/IMU/Barometer data for indoor tracking, and processes MQTT vehicle telemetry.

Tech Stack

Frontend: MapLibre GL Native, Three.js, SceneKit

Backend: MQTT Protocol (Pub-Sub Broker)

Database: SQLite, GeoJSON, IndoorGML

AI/ML, APIs, or Hardware: GPS, Barometer, IMU, VIO Sensor Fusion

Innovation & Uniqueness

- Fully offline-first architecture eliminating cloud and cellular dependencies through pre-compiled ‘.pbf‘ tiles and embedded SQLite schemas.
- Seamless indoor-outdoor transition leveraging smartphone sensor fusion (VIO + Barometer + Map-Matching + BLE).
- Ultra-low bandwidth MQTT telemetry implementation specifically optimized for intermittent IoT campus shuttle tracking.

Feasibility & Scalability

- **Feasibility:** Built using mature open-source frameworks (MapLibre, Three.js, SQLite) requiring standard smartphone hardware sensors.
- **Scalability:** Easily expandable to multi-campus institutions by compiling new OSM spatial bounding boxes and indoor JSON floor graphs.
- **Risks:** Initial manual dimension mapping effort; mitigated by utilizing existing architectural blueprints and automated conversion scripts.

Cost for Product (Optional)

- Development or hardware cost: Only Bluetooth Beacons are the expenses so the cost vary by manufacturer and number of beacons required (approx. Rs.1000/Beacon)
- Hosting or maintenance cost: Minimal local MQTT broker maintenance (Rs.1000/month cloud bridge optional)
- Estimated total: Depends on the number of beacons required, and its manufacturer + Maintenance cost

Use Cases

- Use case 1: Freshers navigating to unfamiliar lecture halls inside complex multi-story blocks like AB1 via indoor 3D floor layers.
- Use case 2: Students tracking live campus shuttle locations in real-time over low-bandwidth connections to optimize transit timing.
- Use case 3: Visiting parents locating administrative offices independently without requiring cellular internet access.

Impacts and Benefits

- Eliminates student stress and academic delays caused by lost lectures and venue confusion.
- Enhances campus accessibility and inclusivity for visitors, guests, and differently-abled individuals.
- Drastically reduces cellular data overhead and battery consumption through efficient client-side vector processing.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Develops resilient, sustainable digital infrastructure for smart campus mobility.
SDG 11	Target 11.2	Sustainable Cities and Communities	Optimizes public transit tracking and inclusive campus accessibility.

Individual Member Contribution

Member	Contribution or responsibility
Vishran S	Found the relevant research papers, topics required to be studied
Harsha Vardhan C	Studied each topic deeply, summarized the final idea
Ram Charan D	Read all the relevant research papers word by word
Lahar B	Read all the relevant research papers word by word

Yukti ID

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